

Practical 2 – JALVIEW exercises

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The P53 gene (and its protein product TP53) is a transcription factor and a master regulator of transcription. **In this exercise, you will explore the evolutionary history of the TP53 protein using Jalview.**

To get started, read about P53 in Wikipedia

<https://en.wikipedia.org/wiki/P53>

1. In Jalview load the PFAM seed alignment of the P53 family. You will use the Fetch Sequences command. The accession ID of P53 is PF00870. You will find additional information needed for these exercises on the UniProtKB web page for P53_human <https://www.uniprot.org/uniprot/P04637>

2. Move the P53_Human protein to the first position in the alignment. Click to select, then use the up arrow key.

3. Color the alignment with “ClustalX” coloring. What amino acid residue properties do the colors represent?

Orange: G (Glycine)

Yellow: P (Proline)

Magenta: D, E (acidic or negatively charged)

Cyan: Y (Tyrosine), H (Histidine)

Blue: F, L, V, W, A, C, I, M, (hydrophobic)

Green: S, T, Q, N (polar or uncharged)

Red: K, R (positively charged)

Pink: C (Cystine)

4. To see the amino acid residue number of a residue, mouse over. The text is displayed in the bar at the bottom of the window. What is the residue number of the S in P53_human in the first column of the alignment.

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5. Do you see examples of columns where a property of the amino acid residues (such as hydrophobicity) is completely conserved, but the column contains more than a single amino acid residue. Do you see columns in which a single amino acid residue is completely conserved?

An example where the amino acid residues are conserved but contain more than a single amino acid residue would be column 55, in which all the residues are hydrophobic but contain amino acids M, L, V, F, and I. An example in which a single amino acid residue is completely conserved is column 105, where all the residues across the rows is C.

6. Locate the columns for the amino acid residues where each of the four zinc-binding sites occurs. What are the amino acids in those positions? Are they completely conserved, almost completely conserved? If there are columns that are almost completely conserved, what species harbor the amino acid that is different from humans?

CYS (176): C, column 105, completely conserved

CYS (238): C, column 215, completely conserved

CYS (242): C, column 219, completely conserved

HIS (179): H, column 108, almost completely conserved

for this column, the species that harbor amino acids different from humans are Q7QBX6 (Anopheles gambiae, African malaria mosquito), W5JCA5 (Anopheles darlingi, mosquito), and A0A084W6W1 (Anopheles sinensis, mosquito)

7. Select five missense mutations that are associated with Li-Fraumeni syndrome. Find the position of each one in the alignment and look at the column. Do you see the mutated amino acid in any of the species in that column? List what the mutation should be. If yes, which species?

R175H: Arginine to Histidine at codon 175

- ARG 175 is located at column 104
- However, there are no species listed here that mutate to His

G245S: Glycine to Serine at codon 245

- GLY 245 is located at column 223
- There are three species that mutate to serine: A0A8R2F7U8 (*Acyrtosiphon pisum*), B7PEY7 (*Ixodes scapularis*), A9V4M3 (*Monosiga brevicollis*)

R248Q: Arginine to Glutamine at codon 248

- ARG 248 is located at column 228
- There is one species that has this mutation: T1EE77 (*Helobdella robusta*)

R273H: Arginine to Histidine at codon 273

- ARG 273 is located at column 254
- There is one species with this mutation: B0W6K4 (*Culex quinquefasciatus*)

R282W: Arginine to Tryptophan at codon 282

- ARG 282 is located at column 263
- But there are no species here that have this particular mutation

8. Select all the sequences in the alignment. Then use the Calculate command to calculate “Tree or PCA”.

a. Select Tree/Neighbor Joining and the BLOSUM62 scoring matrix. Use the output tree to answer these questions:

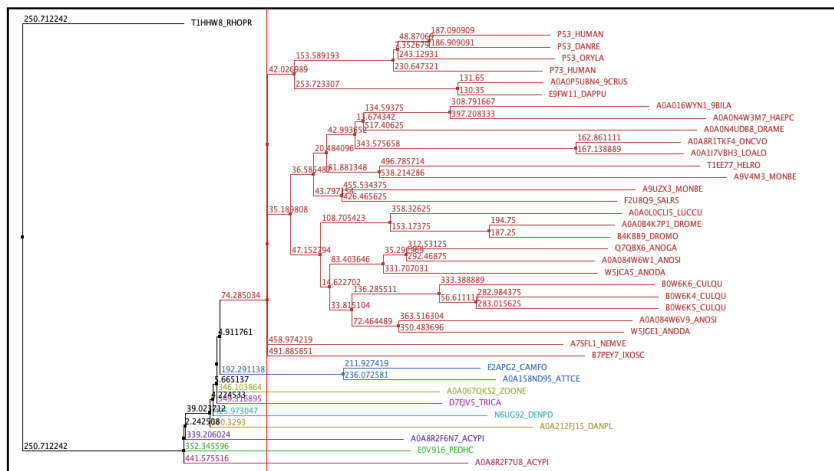
1) What is the closest related P53 protein to human in another species?

P_53 DANRE, Danio rerio (Zebrafish) (Brachydanio rerio)

2) Why is P73_Human more distant from P53_Human than P53_Danre?

From the tree, we can tell that P73 and P53 diverged before the speciation event that separated humans and zebrafish.

3) Click on the tree and move the red line to identify clades of related P53 proteins



To the Right (Clades): You will see the distinct "clusters" or families. At this position, the tree "breaks" into three primary clades:

1. **vertebrates (humans, fish)**
2. **insects (flies, mosquitoes)**
3. **distant clade: contains more divergent sequences**

4) What species harbors the P53-like protein that is the most distant from human P53 on the tree?

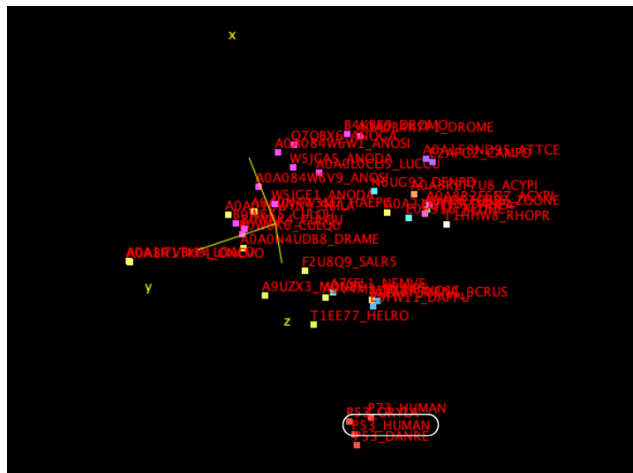
T1HHW8_RHOPR, *Rhodnius prolixus* (Triatomid bug)

It is located at the very top left

b. Select PCA and the BLOSUM62 scoring matrix. Use the output plot to answer these questions.

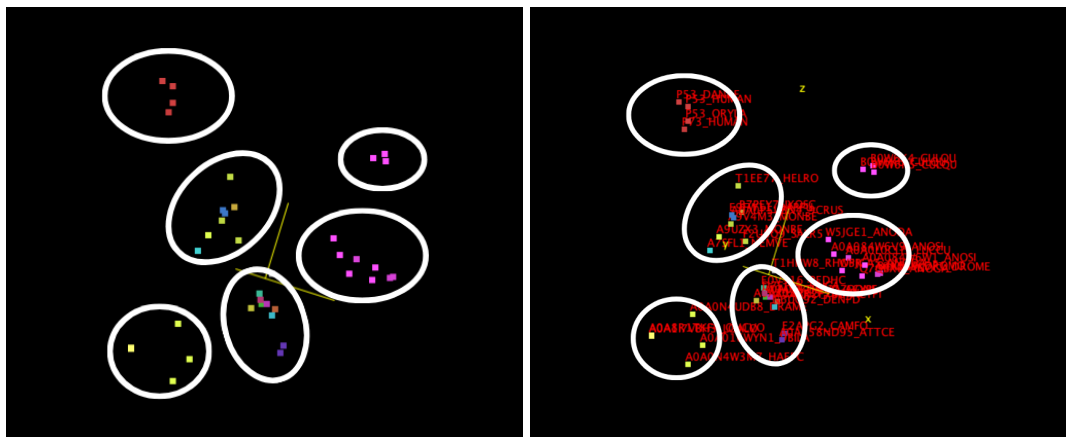
1) View the first three principal components.

2) Find human P53 in the PCA space (HINT: the color should match the P53_HUMAN highlight color in the alignment window).



P53

3) Based on PCA, how many P53 clades are represented in this alignment? 6



Save your Jalview project.

What to turn in: Write the answers to all questions in this document.

Save as p02_<yourjhid>.tar.gz.

Email to fcbbioinf@gmail.com by 11:59PM Feb. 8, 2026